

Unequal Cover: Measuring Privacy Disparities in Proxy Infrastructure

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Abstract

Privacy-focused proxy systems (e.g., multi-party relays, VPNs, etc.) are now standard Internet infrastructure, yet little is known about potential disparities in the privacy they provide across different populations. In this work, we investigate disparities in proxy crowd sizes, measuring how mappings between users and infrastructure can produce unequal geolocation anonymity across communities. We combine infrastructure deployment data with demographic information and apply our framework to Apple's iCloud Private Relay, analyzing 83,609 U.S. census tracts, and find significant disparities: rural users receive crowd sizes over 9× smaller than urban users, and income moderately correlates with privacy outcomes in rural areas ($r=0.39$) while showing negligible correlation in cities. We then explore, through simulation, strategies for improving privacy equity, demonstrating that operators can enhance privacy for the worst-protected populations by up to 2.5× using simple consolidation strategies while introducing manageable potential geolocation error.

CCS Concepts

• Security and privacy → Social aspects of security and privacy; Privacy protections; • Networks → Network measurement.

Keywords

Multi-party relays, Privacy equity, Network measurement, Digital divide

ACM Reference Format:

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1 Introduction

Millions of users [16] now access the Internet via multi-party relay (MPR) systems. These systems aim to protect a user's network identity by routing traffic through two non-colluding (the system

is designed to be operated by separate parties) proxies, such that neither of the two parties has the ability to link user identities to their traffic. However, architectural privacy (i.e., trust separation) does not guarantee that all users receive equal protection in practice.

Beyond masking users' network identities, one privacy metric for proxy systems is the size of the anonymity set (i.e., the number of users who appear indistinguishable to observers monitoring traffic [10]). Larger anonymity sets provide stronger protection by making traffic analysis more difficult. However, there is a tension between maximizing anonymity set sizes and providing a good user experience. For instance, if all users in the United States were mapped to a single egress node, the anonymity set size would be very large, but geolocation accuracy would be poor because all users would appear to connect from the same location. Because of this, commercial MPRs typically map users to egress nodes based on geographic proximity with the goal of maintaining locality and rough geolocation accuracy. This introduces a trade-off: while users may receive more local routing and more accurate geolocation, privacy protections may vary widely across different communities based on infrastructure footprint and demographic factors. In practice, systems like iCloud Private Relay maintain thousands of egress locations across the United States alone, each serving a different slice of the user population. As a result, users assigned to different egresses can see substantially different crowd sizes.

While anonymity properties have been studied in long-running systems like Tor [1, 11], no prior work has examined whether commercial MPR deployments provide equitable protection across communities, despite growing evidence of digital divides in Internet access [13, 14, 17]. MPRs have been studied extensively, but recent work has primarily examined the architecture [19], performance [25], and vulnerability to traffic analysis attacks [27]. To the best of our knowledge, no prior work has studied how privacy outcomes differ across populations in commercial MPR deployments.

In this paper, we present a framework for measuring privacy inequality in proxy systems, using iCloud Private Relay as a case study. We choose Private Relay because Apple's public attestation of egress locations [4] provides transparency that other systems lack, enabling measurement. However, the disparities we identify reflect the interaction between proxy placement, existing infrastructure geography, and population patterns, not properties unique to Private Relay. Combining egress infrastructure data with census demographics [26] across 83,609 U.S. census tracts, we discover systematic disparities: rural, lower-income, and less educated regions experience reduced privacy protections. These factors also



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interact: income is most strongly associated with privacy in rural areas, where incomes and crowd sizes are already lower.

Given these findings, we simulate egress consolidation strategies and show that simple heuristics, such as merging the smallest egresses, can improve privacy for the worst-served populations by up to $2.5\times$ with modest geolocation cost.

2 Background and Threat Model

In this section we describe the proxy architectures relevant to our study and define the privacy property our framework measures.

2.1 Privacy-Preserving Proxy Architectures

Multi-party relay (MPR) architectures split trust across a small number of cooperating intermediaries: an ingress proxy sees user IP addresses but not destinations (due to nested encryption), while an egress proxy sees destinations but not user origins [21]. This split-trust model is used in several deployed systems, most notably Apple’s iCloud Private Relay [3] and Cloudflare’s Warp [9], both of which are built on top of the MASQUE [18, 20] protocol.

Private Relay, our case study, routes Safari and DNS traffic through Apple (ingress) and CDN partners including Cloudflare, Fastly, and Akamai (egress) [21]. Regardless of which CDN handles the traffic, the user’s traffic is mapped to an egress IP address that is shared with other users. These egress IP addresses are distributed across the world, and the user-egress mapping determines the user’s anonymity set. Apple publicly releases a feed of the service’s egress IP addresses and their associated locations [4], listing the cities that IP ranges should be mapped to. This transparency, intended to allow external services to accurately localize Private Relay traffic, also enables measurement studies of the system’s privacy properties.

2.2 Privacy Model and Threat Scope

Proxy systems provide multiple privacy properties. *IP address hiding* conceals the user’s true address from destinations. Systems with multiple exit points may provide *flow unlinkability*, where different connections exit through different addresses. Our framework measures *geolocation anonymity*: the size of the anonymity set of users mapped to the same exit location.

When an external observer resolves an exit IP to a geographic location (using a geolocation database or, in Private Relay’s case, Apple’s public feed [4]), the user is hidden within the “crowd” of all users mapped to that location. However, because users are mapped to egress locations based on their actual location in order to maintain rough geolocation accuracy, the geolocation anonymity varies across users of the system (e.g., a user in New York City may share an exit with hundreds of thousands of potential users, while a user in rural Montana may share one with only tens of thousands). The focus of our work is on this variance.

3 Methodology

We combine egress location data, underlying population demographics, and a configurable mapping function to quantify how anonymity varies across populations under different adoption assumptions.

3.1 Data Sources

Infrastructure Data. We download egress location data from Apple’s public JSON feed [4], which lists IP ranges and their associated geographic locations. As of February 16, 2026, the feed contains 5,925 entries across the United States. Of these, 481 city-state pairs appear more than once (typically as both title-case and all-caps variants), with 85% of duplicate pairs geocoded within 1 km of each other. We merge egresses within 2 km per state using spatial clustering¹, yielding 5,429 unique egress locations. Note that our de-duplication and clustering approach impacts crowd size and anonymity calculations. However, our privacy metric is geolocation anonymity, so merging nearby egresses is a reasonable assumption for an observer trying to resolve user locations.

Demographic Data. We use 2019–2023 American Community Survey (ACS) 5-year estimates from NHGIS [22], extracting total population, median household income, and educational attainment (bachelor’s degree or higher) for each of 83,609 census tracts with valid population data. We use census tracts as our geographic unit of analysis, as they are the smallest unit for which demographic data is available and contain relatively homogeneous populations.

Each census tract is classified by population density: Urban ($\geq 3,000$ persons/mi²), Suburban (1,000–2,999), and Rural ($< 1,000$), as these thresholds capture differences in infrastructure deployment economics. By extension, we refer to an egress, once tracts are mapped to it, as rural, suburban, or urban when the majority of tracts it serves fall into that density class.

3.2 Mapping Models

To conduct our privacy analysis, we model how tracts are assigned to egress locations. Because Private Relay’s actual user-to-egress mapping algorithm is proprietary, we analyze results under multiple plausible mapping models and evaluate robustness (Section 4.3), including:

Nearest-Egress (Baseline). Each tract maps to its geographically nearest egress (using the tract’s centroid) within the same state using great-circle distance. This serves as our baseline, reflecting the assumption that users are assigned to nearby egresses to preserve locality and accurate geolocation by external services. We restrict mappings to the same state given that Apple would likely avoid cross-state routing for regulatory reasons (e.g., live sports streaming rights). Tracts that contain multiple egresses have their population split equally across them. Starting from the 5,429 clustered egress locations, this mapping assigns at least one tract to 5,320 egresses; the remaining clustered locations are not selected by any tract.

K-Nearest Egress ($k=2, k=3$). Each tract distributes its population equally across its k nearest in-state egresses. This models the possibility that Private Relay rotates users across nearby egresses or that different CDN providers serve different egresses for the same geographic area.

State-Constrained Random. Each tract is assigned to a random egress within its state uniformly at random. We run 100 iterations

¹We make the simplifying assumption that an observer resolving egress IPs would map all variants to the same city. Exploration of finer-grained mapping strategies is left for future work.

and report averages. This provides a worst case for our inequality analysis: if random assignment within a state still produces disparities, the finding is driven by infrastructure sparsity rather than our mapping model.

3.3 Privacy Quantification

Adoption Rate Models. Since Private Relay requires a paid iCloud+ subscription, adoption likely varies across demographics. We test multiple adoption models: uniform (0.11% [2], 0.5%, 1%, and 2%); an income-weighted model where adoption scales proportionally with tract median income; and an urban-biased model (2% urban, 1% suburban, 0.5% rural).

We quantify privacy inequality using three complementary metrics:

Crowd Size. The crowd size of an egress is the total population of all tracts (or partial tracts) mapped to it. This estimates the potential user pool behind an egress location, not the exact set of Private Relay users active at a given time. To illustrate: an egress location with a crowd of 370,000 (typical urban median) means geolocation alone narrows a user to a much larger potential population than a crowd of 40,000 (typical rural median).

Degree of Anonymity. Privacy-preserving proxy systems are not used by all users in the crowd, so we measure anonymity using degree of anonymity [10]. We compute $D = \log_2(pN)/\log_2(N)$, where p is the adoption rate and N is the crowd size. This normalizes privacy outcomes relative to population, with values approaching 1 indicating strong relative anonymity. We use $p = .11\%$ as our baseline adoption rate [2] based on reported usage of Private Relay in the United States.

Inequality Index. To quantify privacy disparities between census tracts within regions (e.g., state or county), we compute $I = P_{90}(D)/P_{10}(D)$, the ratio of 90th to 10th percentile degree of anonymity within a region. $I = 1.0$ represents perfect equality; higher values indicate disparities. I allows us to identify regions with high privacy inequality, even in cases where median anonymity may be high. Inequality Index has been used in prior work to quantify disparities in access to cloud infrastructure [14], and we adapt it here to measure disparities in privacy outcomes.

4 Measurement Results

Here we present baseline measurements of privacy inequality, validate their robustness through sensitivity analysis, and then decompose the sources of inequality.

4.1 Population Density and Privacy

Figure 1 shows empirical CDFs of crowd size across urban (35,353 tracts), suburban (18,144 tracts), and rural (30,112 tracts) census tracts. The distributions are clearly separated. Urban tracts have a median crowd size of 369,951, nearly an order of magnitude larger than rural tracts at 39,660 (9.3×), with suburban areas in between (180,946). The gap narrows at the tails: at the 90th percentile the ratio drops to 4.8× (953,977 vs. 200,779), as the largest rural egresses begin to approach suburban levels.

This disparity is consistent with infrastructure density: denser areas place more users behind each egress. The size of the gap,

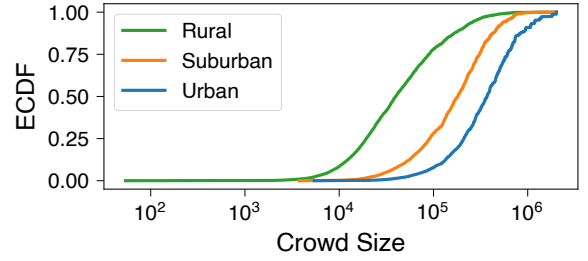


Figure 1: Empirical CDFs of crowd size by population density class.

however, is notable. In practical terms, a 9× crowd-size gap means substantially weaker geolocation anonymity for rural users.

4.2 Socioeconomic Factors and Compounding Disadvantage

We next examine whether socioeconomic characteristics influence privacy outcomes within density categories. Figure 2 shows the relationship between tract median income and crowd size. Income matters most where privacy is worst. In rural areas, the correlation is moderate ($r = 0.39$); in urban areas, it is negligible ($r = 0.06$), with suburban areas in between ($r = 0.25$). We find that educational attainment (bachelor’s degree or higher) shows a similar pattern: $r = 0.29$ in rural areas, $r = 0.21$ suburban, and $r = 0.10$ urban. We omit detailed education plots for brevity.

This yields a compounding disadvantage: in rural areas, privacy is already lower, and income predicts it more strongly there than elsewhere. Rural tracts with median incomes above \$75,000 achieve crowd sizes 2–3× larger than low-income rural tracts, while in urban areas, most tracts achieve large crowd sizes regardless of income. Population density in cities provides a baseline level of privacy for all users, while in rural areas, socioeconomic factors play a larger role in determining privacy outcomes.

4.3 Sensitivity Analysis

We test robustness of our findings by varying user-to-egress mapping and adoption patterns.

Table 1: Mapping model robustness. Findings hold across all “nearest” models.

Model	U/R Ratio	$r_{inc,rural}$	$r_{inc,urban}$	I
Nearest ($k=1$)	9.3×	0.39	0.06	1.57
$k=2$ nearest	8.2×	0.42	0.13	1.55
$k=3$ nearest	7.4×	0.42	0.17	1.54
State-random	1.5×	0.22	0.17	1.24

Mapping Model Robustness. Table 1 shows key metrics under each mapping model. The geographically-grounded models (nearest, $k=2$, $k=3$) consistently show a large urban/rural gap (7–9×), moderate rural income correlation (0.39–0.42), and low urban income correlation (0.06–0.17). Inequality stays high across all three

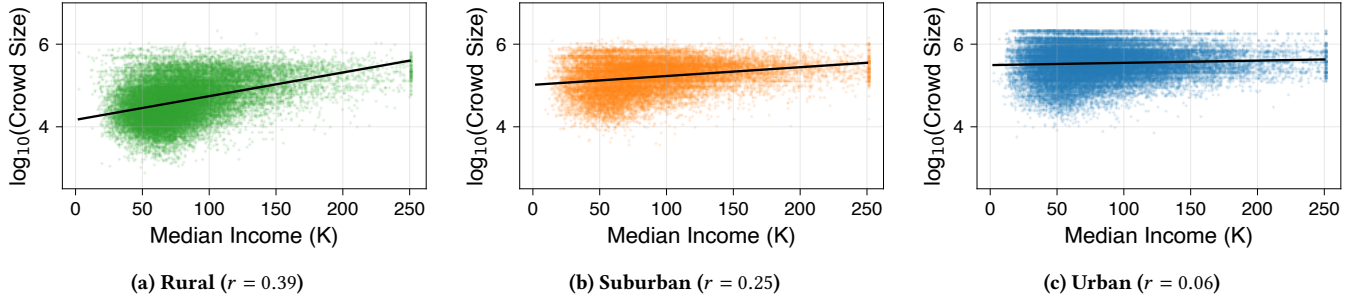


Figure 2: Median household income versus crowd size by density class. Income shows a moderate positive correlation with privacy in rural areas but negligible correlation in urban areas.

models ($I = 1.54$ - 1.57). The mapping choice does not drive our findings. Further, the income correlation *increases* slightly under k -nearest models ($r = 0.42$ at $k=3$ vs. $r = 0.39$ at $k=1$), suggesting the compounding disadvantage pattern is not an artifact of the nearest-egress assumption.

State-random, which destroys geographic locality, shows a reduced but still significant urban/rural gap ($1.5\times$) and dampened income correlations, particularly in rural areas ($r = 0.22$). Even random assignment does not eliminate inequality; rural users still face disadvantages, though the gap shrinks. Further, random assignment would degrade geolocation accuracy and break the locality assumptions operators typically use when assigning egresses, making it unrealistic for real-world use.

Given these results, we focus on the nearest-egress model for the rest of our analysis, as it is realistic, simple, and produces the strongest inequality patterns.

Table 2: Adoption rate robustness. Income-weighted and urban-biased adoption amplify inequality.

Model	$r_{\text{inc,rural}}$	$r_{\text{inc,urban}}$	I
Uniform 0.11%	0.39	0.06	1.57
Uniform 0.5%	0.39	0.06	1.30
Uniform 1.0%	0.39	0.06	1.23
Uniform 2.0%	0.39	0.06	1.17
Income-weighted	0.47	0.21	1.66
Urban-biased	0.41	0.10	1.51

Adoption Rate Robustness. Table 2 shows metrics under different adoption assumptions. Uniform adoption rates shift the overall level of anonymity but preserve the inequality structure. The income-weighted model, a plausible model given Private Relay’s paid iCloud+ requirement, *amplifies* inequality ($I = 1.66$ vs. 1.23 at uniform 1%) and strengthens the income-privacy correlation ($r = 0.47$ in rural areas). In this model, wealthier tracts both have larger crowds (from mapped egress geography) *and* higher adoption (from greater ability to pay), compounding the advantage. The urban-biased model also amplifies inequality, though to a lesser extent ($I = 1.51$ vs. 1.23 at uniform 1%). As discussed, the uniform baseline likely understates real-world inequality.

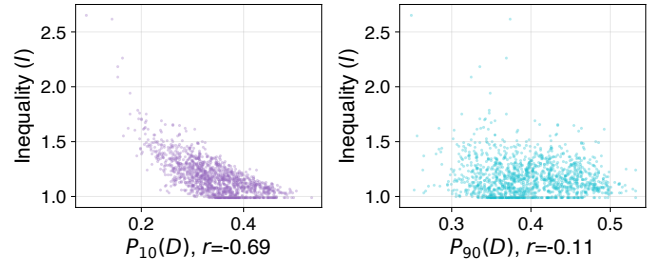


Figure 3: Inequality stems from under-protection of underserved populations.

4.4 Sources of Inequality: Floor vs. Ceiling

We calculate the inequality index I for each county in the U.S. with at least 10 tracts and then analyze how it correlates with the anonymity floor ($P_{10}(D)$) and ceiling ($P_{90}(D)$). Figure 3 shows that inequality strongly correlates with the anonymity floor ($P_{10}(D)$, $r = -0.69$) but shows little correlation with the ceiling ($P_{90}(D)$, $r = -0.11$). Inequality stems from poor protection of underserved populations rather than over-protection of privileged ones.

This asymmetry suggests that the most direct path to reducing inequality is to focus on underserved areas (e.g., increasing egress crowd sizes where the privacy floor is lowest) rather than reducing coverage in well-served areas.

5 Egress Consolidation Simulation

What can MPR operators do about the disparities we observe? Here, we simulate the effect of strategically removing egress locations to improve privacy for underserved populations.

5.1 Simulation Design

We initialize the simulation from the 5,320 mapped egresses and their computed crowd sizes, iteratively removing one egress per step, reassigning its associated tracts to the nearest remaining in-state egress. At each step we record crowd size percentiles (P_{10} , P_{50} , P_{90}), inequality index I , and mean tract-to-egress distance (a proxy for geolocation accuracy degradation).

We evaluate four consolidation strategies:

- **Smallest-first:** Remove the egress with the smallest current crowd, merging the weakest locations first.

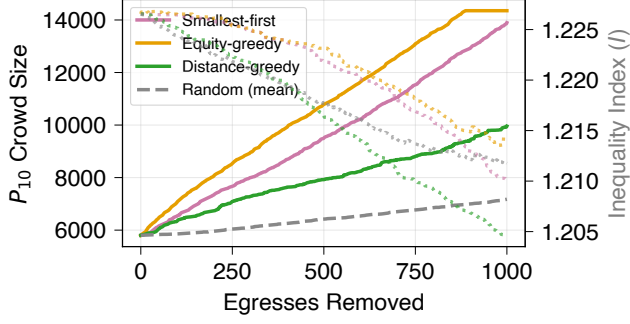


Figure 4: Inequality metrics vs. number of egresses removed. Solid lines show P_{10} crowd size (left axis); dotted lines show inequality index I (right axis).

- **Equity-greedy:** At each step, remove the egress whose removal maximizes P_{10} crowd size, directly targeting the worst-protected populations.
- **Distance-greedy:** Remove the egress whose removal minimizes the increase in mean tract-to-egress distance, preserving geolocation accuracy and locality.
- **Random:** Remove egresses uniformly at random (averaged over 5 runs), providing a baseline for comparison.

5.2 Results

Figure 4 shows how inequality metrics evolve as consolidation proceeds. After removing 1,000 of 5,320 egresses (~19%), smallest-first and equity-greedy both raise P_{10} from 5,800 to roughly 14,000, a 2.5× improvement for the worst-served populations. Distance-greedy optimizes for inequality instead ($I = 1.21$, P_{10} to 10,000); random removal reaches just 7,200.

However, these improvements come at the cost of increased mean tract-to-egress distance, which we use as a proxy for degraded geolocation accuracy. Figure 5 shows this trade-off: equity-greedy increases mean distance from 8.6 km to 10.2 km, while smallest-first incurs slightly less cost (9.1 km). The distance-greedy strategy preserves accuracy best (8.8 km). The curves suggest that the first several hundred removals provide the most benefit per unit of geolocation cost.

We measure only mean tract-to-egress distance as a proxy for geolocation accuracy; consolidation may also affect quality of experience through reduced egress locality (e.g., proximity to CDN infrastructure or regional peering), which our data cannot resolve. Operators with visibility into traffic flows could weigh these effects directly when choosing which egresses to merge.

A tangible result from the simulation is that operators may not need a global optimization step to get most of the benefit. The smallest-first heuristic performs nearly as well as equity-greedy while requiring only local decisions. Removing just 500 egresses (~9%) improves P_{10} by 1.9× with only a 0.6 km increase in mean tract-to-egress distance, a small cost for nearly doubling the privacy floor.

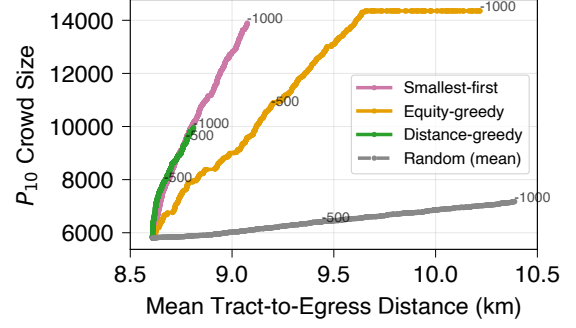


Figure 5: Privacy-accuracy trade-off: P_{10} crowd size vs. mean tract-to-egress distance as egresses are consolidated. Equity-greedy achieves the largest gains for the worst-protected populations.

6 Discussion

Inherited, Not Created. The disparities we measure are not manufactured by Private Relay; they track the existing digital divide. Egress placement follows population density and Internet infrastructure, both of which are already unevenly distributed, so proxy coverage inherits the same geographic and socioeconomic gaps present in the underlying network. What is specific to proxy systems is *amplification*: because iCloud+ adoption likely scales with income, wealthier areas gain privacy both from denser infrastructure and from higher adoption (Table 2), compounding an advantage that infrastructure placement alone would not produce. The practical implication is that equitable privacy will not emerge on its own. Unlike physical build-out, however, the remedy is largely operational: reassigning users across existing egresses (§5) raises the privacy floor without new infrastructure, giving operators agency to compensate for divides they did not create.

Recommendations for Operators. Given our findings, we suggest four practical steps for proxy operators. (1) **Monitor the privacy floor.** The inequality we observe is driven by under-protection of the worst-served populations (§4.4), not by over-protection elsewhere. Tracking P_{10} crowd size across a deployment is cheap and directly identifies the populations most at risk. (2) **Merge small egresses.** As shown in §5, operators can simply merge their smallest egresses into neighbors to achieve significant privacy gains without global optimization. (3) **Account for paid adoption.** Systems with a paid subscription model (like iCloud+) should evaluate non-uniform adoption that may amplify infrastructure-related disparities (Table 2). Uniform-adoption assumptions can underestimate disparities. (4) **Privacy-balance, not just load-balance.** Operators already distribute traffic across infrastructure for performance; they should also consider the privacy implications of that distribution. *Privacy balancing* would spread users across exit points to avoid leaving any population with an inadequately small anonymity set. Our consolidation simulations (§5) show this is feasible: modest reallocation of exit point assignments can substantially raise the privacy floor without requiring new infrastructure.

As proxy-based privacy systems become standardized (MASQUE [20], OHTTP [24]), protocol specifications should

consider minimum crowd size requirements alongside performance targets to ensure that new deployments do not entrench the disparities we observe.

Open Research Questions. Several questions remain open for future research. First, our analysis is cross-sectional; longitudinal measurement of how egress infrastructure evolves over time would reveal whether operators naturally converge toward equitable deployments or whether disparities persist. Second, we measure only geolocation anonymity; combining our measurements with traffic analysis resistance or flow unlinkability would reveal whether geolocation inequality translates into overall privacy inequality. Finally, the compounding disadvantage pattern likely applies to other privacy infrastructure with uneven deployment (e.g., Tor relay placement, encrypted DNS resolver distribution) though this remains untested.

Limitations. We do not know Private Relay’s actual user-to-egress mapping; our sensitivity analysis mitigates but does not eliminate this uncertainty. We also lack traffic volumes, temporal usage patterns, flow-level behavior, and the actual Private Relay user population behind each egress. Our adoption sensitivity analysis shows that these factors affect the magnitude of inequality, and likely make the uniform baseline conservative. Finally, our geolocation privacy metric measures only one aspect of the privacy Private Relay provides; other properties (e.g., flow unlinkability, DNS encryption) are not measured, and census tracts are imperfect proxies for user locations.

7 Related Work

Private Relay Measurement. Prior work on Private Relay has studied its network topology [19], performance characteristics [25], vulnerability to traffic analysis [27], and split-trust architecture [21]. These studies evaluate system-wide properties but do not examine how privacy outcomes differ across populations.

Infrastructure Inequality. A growing body of measurement work shows that infrastructure disparities translate directly into unequal Internet experience. Martin and Dogar [14] found that cloud edge deployments systematically underserve rural and low-income areas, introducing an inequality index we adapt in §3.3. Manda *et al.* [13] showed that targeted investment has not closed them. Bhuiyan *et al.* [5] revealed performance disparities correlated with socioeconomic factors, and Mishra *et al.* [15] showed geographic digital divides in mobile traffic.

Privacy Metrics. Anonymity metrics range from Chaum’s anonymity sets [8], which count indistinguishable users, to Diaz *et al.*’s degree of anonymity [10], which normalizes for non-uniform distributions over the set, and information-theoretic approaches [7, 23]. The extensive Tor measurement literature [1, 6, 11, 12] has studied relay selection, traffic correlation, and performance trade-offs. We repurpose these metrics to measure inequality rather than aggregate security.

8 Conclusion

Our central finding is that proxy placement can reproduce geographic and income-based privacy disparities when it follows existing infrastructure and population patterns. The inequality we measure is driven by under-protection of the worst-served populations rather than over-protection of the best-served. The compounding disadvantage pattern, where income matters most for privacy precisely where resources are scarcest, suggests that proxy operators need explicit privacy-equity goals to compensate for broader digital divides in Internet infrastructure.

As proxy-based privacy becomes standard Internet infrastructure, operators should not just optimize for average anonymity—they should pay close attention to the tail. Our simulation results show that even simple heuristics like merging the smallest egresses can meaningfully raise the privacy floor without significantly changing the infrastructure footprint. A reasonable starting point would be to track minimum crowd-size targets, much as operators already track bandwidth or availability targets.

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